University of BRISTOL

# Introduction and logistics

CADE30008 Flight Dynamics & Advanced Control

Dr. Steve Bullock

# Schedule

| Week | Act | This week | Coursework | Laboratory |
|---|---|---|---|---|
| 1 | | Lecture 1: The design cycle, end to end | | **Quanser** open access |
| 2 | | No control session. Flight dynamics has both of the week's slots. | | |
| 3 | | Tuesday · Lecture 2: Requirements and models you can trust | | |
| 4 | I | Tuesday · Lecture 3: PID, properly | | |
| 5 | | Tuesday · Guest lecture | **Brief released** | |
| 6 | | Consolidation week · no lecture | **Checkpoint 1, provisional** | |
| 7 | | Tuesday · Lecture 4: Stability and margins | | |
| | II | Thursday (TBC) · Lecture 5: Robustness and trade-offs | | |
| 8 | | Lecture 6: Loop shaping | **Checkpoint 2, provisional** | |
| 9 | | Lecture 7: Flight control architecture | **Checkpoint 3, provisional** | |
| 10 | III | Lecture 8: State space and state feedback | | |
| 11 | | Lecture 9: What comes next · then Coursework Q&A | **Due Thursday of week 11** | |
| 12 | | Revision week | | |

Lecture · Guest lecture · Consolidation · No session · Coursework due Thursday of week 11

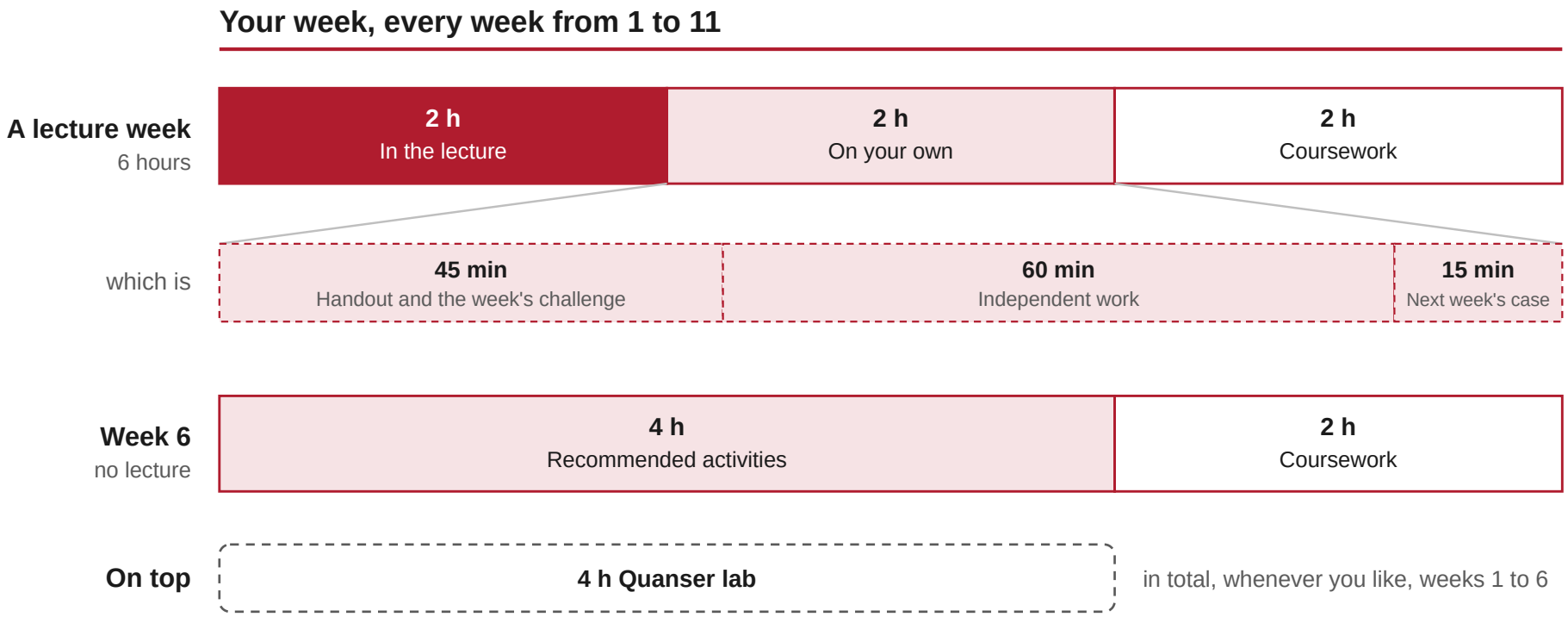
# Assessment

- **The coursework counts for 100% of your grade.** One piece, due Thursday of week 11
- Everything else is **formative** — it doesn't contribute to your grade *directly*
- **But it is how the coursework gets built.** The checkpoints *become* the paper
- **Checkpoints, provisionally weeks 6, 8 and 9.** Automatic checks, a message to the whole cohort, and a personal note for some of you
- Nothing formative is randomised. Work on it together if you like

# Brief release: week 5

- It goes through **external examiner review** first, which is why it is not out on day one
- **Until then, the best preparation is the unit itself.** What you do in and out of these sessions is what the coursework asks for
- **Front-load the laboratory.** It is open access, slots are booked, and the window closes well before the deadline
- **Watch the videos.** Brian Douglas and Steve Brunton, linked from the reading page

# Your week


Your week, every week from 1 to 11
A lecture week
6 hours
2 h
In the lecture
2 h
On your own
2 h
Coursework
which is
45 min
Handout and the week's challenge
60 min
Independent work
15 min
Next week's case
Week 6
no lecture
4 h
Recommended activities
2 h
Coursework
On top
4 h Quanser lab
in total, whenever you like, weeks 1 to 6

# The Quanser laboratory



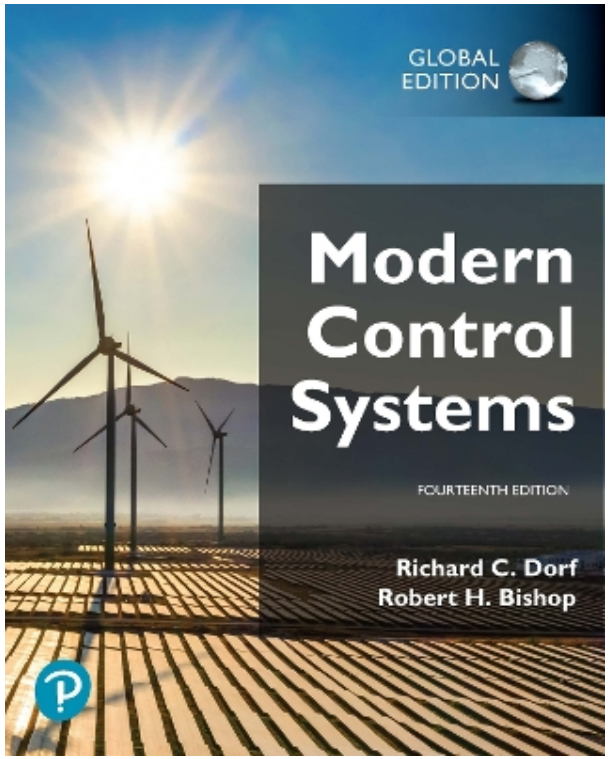

**Where** M.003, under the Queens Building cafe, with the flight simulators

**Booking** Link on Blackboard.

**Who with** Pairs or threes, one of you books

**Four hours, open access until end of week 6.**

# One book worth knowing about




**Dorf and Bishop, *Modern Control Systems***
— Pearson, any edition from the 12th

- **You do not have to buy it.** The handouts are the authoritative version and stand on their own
- Closest in spirit to this unit: it frames everything around **the design cycle**
- Use it for **a second explanation** to complement these resources, and for **more worked examples**
- The library has it. Which sections go with which week is on the reading page

# Share the library copies

**20 print copies and 14 eBook licences, for about 190 of you**

*The 12th, 13th and 14th editions added together, not copies of one edition.*

- Reading an eBook online locks that copy while you have it open.
- Downloading the whole book locks a copy for 24 hours.
- Downloading a PDF chapter locks nothing, but is capped at 56 pages.

**Download the chapter you want and get out.**

# AI in this unit

- **AI rewards expertise.** Anyone can produce passable-looking work, until it comes up against an expert *or the real world*
- **Work, or gym?** If only the result matters it's work. If *doing* it is the point, it's the gym

> A design in a domain you don't understand, produced by a tool you don't understand and can't check, is **not responsible, not ethical, not engineering**.

# What you're here to build

**Judgement:** knowing a plausible answer is wrong, before you can say why.
**Taste:** knowing which of several correct designs is the good one.

**Nobody can hand you either — not me, not a model. Do the hard part.**